

DESIGN EXERCISE 1

RE-DESIGNING THE LIFT CONTROL UI

Our first design exercise is to take the ‘grouping-separating-prioritising-sharpening’ rule (and any other we can think of) and apply it to a common day-to-day UI, lift control panel. This is a trio project (i.e. 3 people in a group).

DESCRIPTION

There are so many poorly-designed elevator controls... the arrangement of the buttons and how they are grouped/ separated/ prioritized/ sharpened makes a difference in its usage. For example, is it difficult to find the floor button? When the door was closing, it was difficult to quickly find the ‘open’ button and press? What will be an ideal arrangement of the buttons?

This design exercise will help you think about these, particularly in terms of applying the grouping-separating-prioritizing-sharpening which we learned during the lecture.



Other than separating the buttons based on their functions (e.g. floor buttons vs. close/open buttons, or above ground floor vs. below ground floor, etc.), there are other ways to improve the arrangements. For example, one of the most important design principles (which will be summarised later in the module) is called “mapping”: to map the physical world to virtual (UI) world. Use of this principle in the re-design of the lift

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can make the designed UI more “intuitive” in that how it is normally perceived in real world can be used to operate how the UI is.



Design by Taewon Hwang, Hyundai Asan, 2011

For example, the design of light switches on the right demonstrates a direct application of the “mapping” principle in which the spatial floor partitions were mapped to the UI so that there won’t be repeated turning on/off of buttons to see which one turns which partition.

Similar mapping can be applied to lift controls where the buttons for higher floors can be located higher on the panel.

Assume there are 15 floors + ground floor, and also 4 levels underground in the building. Sketch the lift control panel as you apply these concepts, and as you do, you will see how it might be further revised, so re-sketch it and see the result. Find any literature (papers, books, etc.) to help you ideate and also to support your argument. Make sure all sources used for ideation be listed. The final design and your rationale should be your main submission inside the 3-page report. Earlier sketches can be added as appendix, and any literature used to support your design should be listed in the reference, again outside of the 3-page limit.

SUBMISSION

3 pages excluding title page, table of contents, references and appendices, in PDF format. Late submissions will be penalized by 20% deduction of grade per day.

Deadline: 11:59pm, 4th October 2026

USEFUL LINKS

Bartholdi III, J. Elevator control panels. Georgia Institute of Technology (accessed August 2021). <https://www2.isye.gatech.edu/~jjb/misc/elevators/elevators.html>

Shravya, S. Analysing elevator controls using Nielsen-Norman’s usability heuristics, 2019. <https://uxdesign.cc/analyzing-elevator-controls-using-nielsen-normans-usability-heuristics-53e385fa8003>

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The submission constitutes 15% of the module grade.

GRADING RUBRICS

Use the following rubrics to strengthen the submission

Criteria	Marks
Reasonable and sensible ideation	0 - 3
Quality of design (considerations in sketch details, applying guidelines/principles/concepts learned)	0 - 3
Literature and/or other hard evidences to support the ideas and design	0 - 3
Evidence of refinement/evolution in design	0 - 3
Presentation (writing, illustration, etc.)	0 - 3